

Introduction to Databases

(INFR10080)

(Propositional Logic)

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Changelog

v25.0 Initial version

Building blocks

Propositions

Atomic statements that cannot be further decomposed

For example:

- “It is raining”
- “The cat is on the table”
- “The sky is blue”

Usually denoted with uppercase letters: $P, Q, \dots$

Also called **propositional variables**

Logical connectives

- Conjunction: $\wedge$ (**and**)
- Negation: $\neg$ (**not**)

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Syntax

Let **Prop** be a countable set of propositional variables

The **language** $\mathcal{L}$ of propositional logic over **Prop** is defined inductively as follows:

1. Every $P \in \mathbf{Prop}$ is in $\mathcal{L}$ (**atomic formulae** or **atoms**)
2. If ϕ and ψ are in $\mathcal{L}$, then $\phi \wedge \psi$ is also in $\mathcal{L}$
3. If ϕ is in $\mathcal{L}$, then also $\neg\phi$ is in $\mathcal{L}$
4. Nothing else is in $\mathcal{L}$

In other words, $\mathcal{L}$ is generated by the following grammar:

$$\phi := P \mid \phi \wedge \phi \mid \neg\phi \quad (\text{with } P \in \mathbf{Prop})$$

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Semantics

Intuition

- Atomic statements can be either **true** (t) or **false** (f)
- The truth value of a formula is determined by the truth values of its atoms

Formally

A **truth-value assignment** is a function $\alpha: \mathbf{Prop} \rightarrow \{\mathbf{t}, \mathbf{f}\}$

Then, α **satisfies** a formula ϕ (written $\alpha \models \phi$) inductively as follows:

$$\begin{array}{lll} \alpha \models P & \iff & \alpha(P) = \mathbf{t} \\ \alpha \models \neg\phi & \iff & \alpha \not\models \phi \\ \alpha \models \phi \wedge \psi & \iff & \alpha \models \phi \text{ and } \alpha \models \psi \end{array}$$

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Truth tables

Reflect the semantics of the connectives

$\wedge$	t	f
t	t	f
f	f	f

	$\neg$
t	f
f	t

Given α and ϕ , allow us to determine whether $\alpha \models \phi$:

1. Replace each propositional variable P in ϕ by $\alpha(P)$
2. Propagate **t** and **f** according to the truth tables

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Example

Given the formula

$$\phi = \neg(\neg A \wedge B) \wedge \neg(C \wedge \neg D)$$

and the assignment

$$\alpha = \{ A \mapsto \mathbf{t}, \quad B \mapsto \mathbf{f}, \quad C \mapsto \mathbf{f}, \quad D \mapsto \mathbf{t} \}$$

does α satisfy ϕ ?

Yes

Given α and ϕ , checking whether $\alpha \models \phi$
can be done in **polynomial time** in the size of ϕ

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Satisfiability and Validity

A formula ϕ is

satisfiable if there is some α that satisfies ϕ

unsatisfiable if ϕ is not satisfiable

falsifiable if there is some α that does not satisfy ϕ

valid if every α satisfies ϕ
(in this case ϕ is called a **tautology**)

Consequences

- ϕ is a tautology iff $\neg\phi$ is unsatisfiable
- ϕ is unsatisfiable iff $\neg\phi$ is a tautology

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Equivalence

Two formulas are **logically equivalent** (written $\phi \equiv \psi$) if for all α

$$\alpha \models \phi \iff \alpha \models \psi$$

(i.e., there is no assignment that satisfies one formula but not the other)

Intuition: Equivalent formulae have the **same meaning** even though they may **differ syntactically** (they say the same thing in different ways)

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Propositional logic: Extended language

Syntax

New connectives:

- Disjunction $\vee$ (**or**)
- Implication $\rightarrow$ (**if-then**)
- Double implication $\leftrightarrow$ (**if and only if**)

New grammar:

$$\phi := P \mid \neg\phi \mid \phi \wedge \phi \mid \phi \vee \phi \mid \phi \rightarrow \phi \mid \phi \leftrightarrow \phi$$

Semantics

$\alpha \models \phi \vee \psi$	iff	$\alpha \models \phi$ or $\alpha \models \psi$
$\alpha \models \phi \rightarrow \psi$	iff	if $\alpha \models \phi$, then $\alpha \models \psi$
$\alpha \models \phi \leftrightarrow \psi$	iff	($\alpha \models \phi$ if and only if $\alpha \models \psi$)

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Expressive power

Every formula in the extended language
can be **equivalently expressed** using only $\wedge$ and $\neg$

- $\phi \vee \psi \equiv \neg(\neg\phi \wedge \neg\psi)$
- $\phi \rightarrow \psi \equiv \neg\phi \vee \psi$
- $\phi \leftrightarrow \psi \equiv (\phi \rightarrow \psi) \wedge (\psi \rightarrow \phi)$

The new connectives do not add **expressive power** to the language

- they are just **syntactic sugar**
- but useful to write formulae more succinctly

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Truth tables of the new connectives

Disjunction

$\vee$	t	f
t	t	t
f	t	f

Implication

$\rightarrow$	t	f
t	t	f
f	t	t

Double Implication

$\leftrightarrow$	t	f
t	t	f
f	f	t

All of the above can be derived from the truth tables for $\wedge$ and $\neg$

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Equivalences

Commutativity

$$\phi \vee \psi \equiv \psi \vee \phi$$

$$\phi \wedge \psi \equiv \psi \wedge \phi$$

Associativity

$$(\phi \vee \psi) \vee \chi \equiv \phi \vee (\psi \vee \chi)$$

$$(\phi \wedge \psi) \wedge \chi \equiv \phi \wedge (\psi \wedge \chi)$$

Distributivity

$$\phi \wedge (\psi \vee \chi) \equiv (\phi \wedge \psi) \vee (\phi \wedge \chi)$$

$$\phi \vee (\psi \wedge \chi) \equiv (\phi \vee \psi) \wedge (\phi \vee \chi)$$

Idempotence

$$\phi \vee \phi \equiv \phi$$

$$\phi \wedge \phi \equiv \phi$$

Absorption

$$\phi \vee (\phi \wedge \psi) \equiv \phi$$

$$\phi \wedge (\phi \vee \psi) \equiv \phi$$

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Equivalences (continued)

Double Negation

$$\neg\neg\phi \equiv \phi$$

De Morgan

$$\neg(\phi \vee \psi) \equiv \neg\phi \wedge \neg\psi$$

$$\neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi$$

Implication

$$\phi \rightarrow \psi \equiv \neg\phi \vee \psi$$

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Entailment

We extend the **satisfaction relationship** $\models$ to **sets Σ of formulae**:

$$\alpha \models \Sigma \iff \alpha \models \phi \text{ for all } \phi \in \Sigma \iff \alpha \models \bigwedge_{\phi \in \Sigma} \phi$$

Then, we say that Σ **entails** a formula ϕ if for all α

$$\alpha \models \phi \text{ whenever } \alpha \models \Sigma$$

(i.e., every assignment that satisfies Σ also satisfies ϕ)

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Properties of Entailment

Deduction Theorem

$$\Sigma \cup \{\phi\} \models \psi \text{ iff } \Sigma \models \phi \rightarrow \psi$$

Contraposition Theorem

$$\Sigma \cup \{\phi\} \models \neg\psi \text{ iff } \Sigma \cup \{\psi\} \models \neg\phi$$

Contradiction Theorem

$$\Sigma \cup \{\phi\} \text{ is unsatisfiable iff } \Sigma \models \neg\phi$$

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Decision problems

A **decision problem**

- takes (one or more) inputs
- answers a Yes / No question

Example

SAT: Given a propositional formula ϕ , is ϕ satisfiable?
(i.e., can we find an assignment α such that $\alpha \models \phi$?)

A **decision procedure** is an algorithm that

- always **terminates**
- **solves** a decision problem

A decision problem is **decidable** if there is decision procedure for it

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Solving SAT

Satisfiability in propositional logic is a **decidable** problem

Naive algorithm

1. Enumerate all possible assignments
(there are 2^n where n is the number of atoms in the formula)
2. For each assignment check whether the formula is satisfied:
 - 2.1 if it is, stop and answer YES
 - 2.2 otherwise, continue to next assignment
3. If there are no more assignment, stop and answer NO

SAT is an **NP-complete** problem:

- a (candidate) solution can be verified in polynomial time
- no known efficient (polynomial) way to locate a solution
(in the worst case, it requires **exponential** time)

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Reduction to satisfiability

Validity, equivalence, and entailment

can all be reduced to checking satisfiability:

- ϕ is valid iff $\neg\phi$ is not satisfiable
- $\phi \models \psi$ iff $\phi \rightarrow \psi$ is valid (deduction theorem)
- $\phi \equiv \psi$ iff $\phi \leftrightarrow \psi$ is valid

A decision procedure for satisfiability is all we need

Validity, equivalence and entailment are decidable problems
(in particular, **coNP-complete**) in propositional logic

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Some additional terminology

Atom atomic formula

Literal atom (**positive literal**)
or its negation (**negative literal**)

Clause disjunction of literals

Term conjunction of literals

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Normal forms

Formulae expressed in a standard syntactic form

Conjunctive Normal Form (CNF)

Conjunction of clauses: $\bigwedge_{i=1}^n (\bigvee_{j=1}^m L_{i,j})$

Example: $(A \vee \neg B) \wedge (B \vee \neg C \vee \neg D)$

Disjunctive Normal Form (DNF)

Disjunction of terms: $\bigvee_{i=1}^n (\bigwedge_{j=1}^m L_{i,j})$

Example: $(A \wedge B) \vee (A \wedge \neg C) \vee (A \wedge \neg D) \vee (\neg B \wedge \neg C) \vee (\neg B \wedge \neg D)$

Negation Normal Form (NNF)

Only $\wedge$, $\vee$, $\neg$ and negation can appear only in front of atoms
(in particular, every formula in CNF or DNF is also in NNF)

Example: $(A \wedge (B \vee \neg C \vee \neg D)) \vee (\neg B \wedge (B \vee \neg C \vee \neg D))$

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Converting to NNF, CNF and DNF

For every formula
there exist **equivalent** formulae in **CNF**, **DNF** and **NNF**

To convert into NNF

1. Replace $\{\rightarrow, \leftrightarrow\}$ by $\{\wedge, \vee, \neg\}$
2. Apply De Morgan's laws

To convert into CNF or DNF

1. Convert into NNF
2. Apply the distributivity laws

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Why normal forms?

Some approaches to inference use syntactic operations on formulae, often expressed in standardized forms

CNF tells us something as to whether a formula is **valid**:

- If all clauses contain complementary literals, then the formula is a tautology
- Otherwise, the formula is falsifiable

DNF tells us something as to whether a formula is **satisfiable**:

- If all terms contain complementary literals, then the formula is unsatisfiable
- Otherwise, the formula is satisfiable

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DNF and Satisfiability

Whether a formula in DNF is satisfiable can be decided in **linear time** in the size of the formula

Wait! Didn't we say that SAT is NP-complete? **Yes**

So what's the catch with DNF?

The transformation into DNF is **expensive** (in time/space)

- The size of the formula may **blow-up exponentially!**

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